

A PRIMER REAL ESTATE PROJECT FEASIBILITY STUDY ON AUTOMATED CAR PARKING SYSTEM IN CEBU CITY

PROJECT BACKGROUND AND HISTORY

I.

PROJECT BACKGROUND

A. Project Data

A.1. Description

This Project Feasibility Study will mainly focus on real properties with potentiality and viability for investment wherein the same may be utilized as car parks (the study) held out for lease, thereby yielding profits out of the paid rents and at the same time bestowing convenience to the public.

The project data required for this study will mostly come from researches of the countries like the United States and China who innovatively engineered the latter-day car park system commonly known as the **Automated Car Parking System (APS)**.

The investment and development cost required for the study may not be accurately approximated due to the cost and time limitation involved, hence, the necessity to resort to valid and logical assumptions that from time to time, may be revealed here in this study.

B. Major Project Parameters

The major project parameters that will served as the guiding principles in the preparation of this study are as follows: **(1) Time; (2) Cost; (3) Quantity and; (4) Quality.**

B.1. Time

In the conduct of this study, time element is of the essence.

As earlier relayed, for the investment and development cost required for the study to be accurately approximated, the herein sole consultant or even any other consultant or project team for the matter, necessarily requires time affluence.

However, with the time constraint present, this study will be conducted at a minimum of two (2) weeks and a maximum of one (1) month without sacrificing reliability over compliance.

So also, since this study is self-initiated, there are no relationship, whatsoever about the budget and the deadline because the same is not conducted for some investor who may very well invest over a particular period of the year.

Lastly, for the fulfillment of the study, herein consultant had previously considered considerable numbers of critical time points which are ought to be met.

B.2. COST

Drawing up the budget for the conduct of this study is not that much of a concern, rather, herein consultant is of the position that Human Capital,

(i.e., an individual's knowledge, skills and expertise on the subject matter) is what is more of a pressing matter.

Here, as there is no budget involved in the conduct of this study, but only Human Capital, budget overspending therefor is a "*non-sequitur*".

However, income as a result of the pursuance of this study ought to be accounted for accordingly. Anyhow, topics relative to income requirement, generation and allocation will be more extensively discuss in the **Financial Study**.

B.3. QUANTITY

Quantity is one of the important factors to consider for this study simply because the project aims to cater the public, in general, and private utility vehicle owners, in particular.

Therefore, herein consultant, likewise finds it compelling to know the number of the market, specifically the approximate number of private utility vehicle owners going to and fro the City that would most likely avail of the said facility.

B.4. QUALITY

Finally, awareness of the quality parameters involved in the project, too, is a must have factor. Quality factor is further subdivided into four aspects, as follows:

- (1) Inputs- Involves the raw materials for the project;
- (2) Outputs- Considers what the project produces;
- (3) Outcomes- Considers the results of the project and;
- (4) Processes- Determines the means used to achieve the outputs.

Considering these quality factors in the study, there will be quality factor in the materials that will be used for the automated car parking system (the input), in the delivery of the basic services of pay parking (the

process), in the number of clientele availing of the said services (the output), and in the benefits that the business brought to the public (the outcomes).

Ideally, the four project parameters must concur with each other, however, such a demanding requisite will hardly exist. Sometimes, one parameter need be given more importance than the other. For example, in this study, quality of the services of the automated car parking system and number of the target market that will be catered with the said business far more outweighs the cost and the deadline. Withal, it bears stressing that herein consultant readily identified what the most important parameters are.

C. POLICIES

Considering existing policies in the conduct of a feasibility study is an indispensable step to take. Knowing these relevant existing policies prevents herein consultant's time, effort and resources in going to waste.

These existing policies is a tool needed to know a study's limitation and restrictions. So also, with the limitations and restrictions of the consequent conduct of the operations.

The policies supposedly to be considered in the conduct of the study are as follows: (1) Economic Policy; (2) Industrial Policy; (3) Financial Policy; (4) Social Policy and; (5) Other related policies.

However, the herein consultant being learned in law, the latter deemed it appropriate to discuss the Legal Policies involve for the propose study, rather than the herein preceding policies.

C.1. LEGAL POLICIES

Knowing the legal policies involved in the study, will require one to first consult the fundamental law provisions about land acquisitions.

During one's study of the constitutional provisions as regards land acquisitions, one will readily be informed of the following restrictions:

1. That, only Filipinos may acquire land/s in the Philippines. Except through inheritance;
2. That, the use of marine wealth is exclusive to Filipinos;
3. That, in co-production, joint venture, or production sharing for the exploration, development and utilization of natural resources, foreign equity in a corporation is limited to 40% and;
4. For public utilities, a corporation must be controlled by at least 60% by Filipinos and 40% by foreigners.

Nonetheless, relating these mentioned fundamental restrictions about land acquisitions to this investigation will likewise leads us to conclusion that only items nos. 1 (one) and 4 (four) finds relevant applications.

Aside from the constitutional restrictions as regards land acquisition, a statute of a subsequent legislation is also worth to mention. The Investors Lease Act (R.A. 7652) somehow remedied the prohibition of foreigners' land acquisition by allowing them instead to lease lands for a period of fifty (50) years, renewable for another twenty-five (25) years.

The purpose of allowing the lease is for the obvious reason of allowing foreign investors to pursue their business endeavors in the Philippines conditioned upon the following: (1) Investment must not be less than Five Million Pesos (PHP5, 000,000.00) and (2) the leased area must be for long-term lease "as may be reasonably required for the purpose."

Having these legal policies in mind, as regards land acquisitions and restrictions, the consultant, at the outset directly determine feasibility of the propose study.

II.

PROJECT PRINCIPAL AND/OR INITIATOR

A. NAME/S AND ADDRESS/ES

As this is a study conducted for purposes of compliance to a school requirement, the identity of the consultant and the project principal and/or initiator lies on one and of the same investigator.

PROJECT PRINCIPAL AND/OR INITIATOR	
NAME	ADDRESS
BELTESHAZZAR L. CABACANG	BLASAB, Poog, Toledo City, Cebu, Philippines 6000

B. FINANCIAL POSSIBILITIES

Once more, this study being self-initiated, solicitation of financial plans might be of remote possibility. But, herein consultant does not discount the fact that investor/s in the future might come across this study and may likewise be interested investing in the latter-day parking system, the Automated Car Parking System.

Therefore, this study will not be dwelling much of the financial plans called for.

Nevertheless, one necessary corollary aspect of financial possibility, aside from financial plan is the financial opportunities. While herein consultant will not hold forth much of the financial plans, financial opportunities is but of a quintessential concern.

Hence, herein consultant or any other advisor for the matter, finds it of a paramount duty to discern the financial opportunities that will be brought by the study.

Thus, to comply with the said duty, hereinbelow provided, are some of the tangible financial opportunities relative to the study, as follows:

- 1) International companies with indispensable responsiveness in furnishing technological services will be very much interested to invest in the business;
- 2) National and local governments, in its proprietary capacity will invest in the said specialty for the twin purposes of public service and proprietary interest.
- 3) Local Condominium Developers might adopt said parking system requiring a minimum amount of space, thereby utilizing the other free spaces for other profit earning purposes;
- 4) Car companies too, may soon engaged in the like business not just to have a complex car parking system, but also to effectively display the same.

C. ROLE WITHIN THE PROJECT

Herein consultant's role in the project runs from being an all-knowing researcher down to being the messenger.

As stated elsewhere, this study is to be submitted as a fulfillment of one of the requirements for the Degree of Bachelor's in Science Major in Real Estate Management.

Consequently, not having someone to rely for the success of this study, a writer's communication skills, a financial analyst's analytical skills, an engineer's technical skills, an architect's creative and imaginative skills, an agent's interpersonal skills, and a lawyer's skill of arriving into sound judgment are mandatory skills required for the accomplishment of this undertaking.

III. PROJECT HISTORY

A. HISTORICAL DEVELOPMENT OF THE PROJECT (DATES OF ESSENTIAL EVENTS IN PROJECT HISTORY)

The subject Real Estate Investment and Consulting, being offered during the summer results to a narrow opportunity for the conduct of an exhaustive research or project feasibility study.

Thereupon, the birth of the project was rooted at the very first day of the summer class and will be culminated amid the said duration.

For this reason, every day of the given time, is historical, if not essential.

B. STUDIES AND INVESTIGATION ALREADY PERFORMED.

Before embarking in this schoolwork, the complete fundamental knowledge for the conduct of a feasibility study was previously premeditated.

Topics which encompasses Project Feasibility Study, Investment Measurement Tools, Real Estate Market Analysis, Market Driven Strategy, Highest and Best Use Analysis, Location Analysis, Investment Analysis for Proposed Projects, among others, are discussed and very well deliberated.

REAL ESTATE PROJECT FEASIBILITY STUDY

I. PROJECT SUMMARY

A. NAME OF THE PROJECT & LOCATION

This project is dubbed "**Real Estate Project Feasibility Study on Automated Car Parking System in Cebu**", it being a study undertaken as requirement under the course Bachelor of Science in Real Estate Management at the University of Cebu.

So also, brought about by the demand of the mobility of today' time, comes also the demand for the maximum use of parking spaces.

B. DESCRIPTIVE DEFINITION OF THE PROJECT

Herein consultant choose the project not only for purposes of earning profits, but also to contribute to the benefit of the riding public through the eventual realization and implementation of this disquisition.

Too, aside from this dual intention, herein consultant is also with the ambition of providing the target market with warm, convenient, and relax "Automated Car Parking System" experience.

C. PROJECT OBJECTIVE

Given the demand in transportation and mobility, specifically here in Cebu City, it is the consultant's aim to savvy whether this feasibility study will be

of much help to the society in providing convenient and easy access parking methodology.

So, herein consultant supposedly considers various project objectives, but for the same to be realistic, the latter deems to streamline the project prospective, to wit:

- A. To introduce the Automated Parking System to the Philippines, specifically in Cebu City and to benefit from it, both proprietarily and from the convenience, comfort and welfare it may bring;
- B. To make a comparison of the various aspects of the existing parking system with the Automated Parking System;
- C. To determine the economic benefits that it may brought to the Country, in general and to the City, in particular.

D. FEASIBILITY CRITERIA

The following are the feasibility criteria that the herein consultant wish to prove during the course of this study, to wit:

- (1) That Automated Car Parking System, being a novel commodity will bring forth investors willing to invest in the Country;
- (2) That aside from investors willing to invest in the Country, several patronizers will most likely avail of the services with warm, convenient and relax "Automated Car Parking System" experience;
- (3) That because of several patronizers availing of the said services, demands exists to support companies entering in these industries and;
- (4) That because of several companies entering in these industries, revenue projections will likewise be accomplished.

E. LIMITING CONDITIONS

The study sampled only a limited number of vehicle owners, a potential investor and few officers in public sector. This is because, herein consultant is faced with time constraints that he must intelligently allocate.

Correspondingly, the generalization of the result cannot easily be justified.

II. MARKET STUDY

A. PROJECT DESCRIPTION (DEVELOPMENT THEME)

Our country, being a developing country, has inevitably developed overtime. Also, given the present government's "build, build, build" program, many major infrastructures, including contacted roads had been built, which in turn lead to the increasing number of automobiles.

Consequently, this daily increase of number of automobiles likewise command an increase in the number of parking spaces demands.

Presently, it is of public knowledge that the Philippines still adopts the manual procedure of parking. This kind of parking sometimes exhibits one's lack of discipline. Here, a driver would usually just park anywhere, in a place that convicts him. As such, a huge traffic situation usually occurs in that station.

Besides, car dents are obtained from parking and retrieval due to lack of sufficient parking space and interval.

These circumstances lead to arguments which brings forth chaos, waste of time caused by traffic jam, economical loss consequent of repairs and loss of probable profits from other affairs.



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Portrait 1. Existing Car Parking

Aside from economic loss due to an individual, the same economic loss will likewise be suffered by the business sector, in principle. This economic laggard is due to the fact that business centers, at present, usually have parking hazards.

Taking all these matters into account, herein consultant finds it necessary to find a solution that can overcome these problems that surmount. The "Automated Car Parking System" is the key that herein consultants presents as the solution to these problems.

Over and again, this system is not only for purposes of profit making, but also a project which is ground breaking.



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Portrait 1.1: Proposed Automated Car Parking System

B. FACTORS AFFECTING THE MARKET

Factors affecting the market varies, primarily from internal factors to external factors. It is known that only the internal factors can be actively and directly controlled through marketing management. Nevertheless, there are some external factors that can be overcome by discerning marketing and advertising.

B.1. ECONOMIC FACTOR

The Automated Car Parking System will surely deliver economic benefits which are unparalleled and without skepticism.

B.1.1. Maximum Land Use

An apparent physical characteristic of APS is the use of less area and volume of realty spaces than the other car parking places.

Said physical characteristics therefore results to Maximum Land Use where tenants or developers considers various options such as: the minimization of the land exploitation for the maximization of car parking stations.

B.1.2 Potential Profitability

Automated Car Parking System brings potential profitability even to lands in distressed and of futility.

The design of the system, being flexible, it converts existing land use profitable because APS can be installed inside, under or between irregular structures unnoticeable.

B.1.3 Minimal Maintaining Cost

Unlike in conventional car parks, Automated Car Parking System which is usually located indoors, requires no or minimal staff and security, clean up, monitoring, fire suppression, lighting and ventilation.

B.1.4. Lower Occurrence of Insurable Risks

Insurance Companies usually assumes risk on the probability of accidents or occurring events. The happening of the insurable risks such as theft, property damage, injury or death will less likely become brisk.

B.1.5. Tax Deductions

The investors or property developers may avail of tax deduction from one's total gross income for depreciation costs, thereby lessening

one's total tax liability compared to investors or property developer of traditional car park system.

III. TECHNICAL STUDY

A. PROPOSED IDEAL PROJECT AND CONSTRUCTION SCHEDULE

Calendar for 2018			
May	Jun	Jul	Aug
• Determining the objective of the Project • Conduct researches related to the Project • Collecting ideas for physical structure and technical designs by technical team			

Calendar for 2018			
Sep	Oct	Nov	Dec
• Budget panning and outlining • Gathering of materials • Completion physical structure and technical designs prepared by technical team			

Calendar for 2019			
Jan	Feb	Mar	Apr
• Actual works on physical structure and technical designs by a technical team			

Calendar for 2019			
May	Jun	Jul	Aug
• Presentation			

B. SAMPLE CONSTRUCTION COSTS

Construction / Development Costs Conventional Garage vs Robotic Parking Systems¹

Construction / Development Cost	Conventional Garage	Robotic Parking System
Land (or attributed cost of area)	\$10 Million	\$10 Million
Construction Costs Turnkey	400 spaces at \$20,000 = \$8 Million	850 spaces at \$25,000 = \$21.25 Million
Soft Costs	\$1 Million	\$1 Million
Development Costs (Land + Construction Costs + Soft Costs / # Cars)	\$19 Million/400 = \$47,500/space 125% = \$47,500 / space	\$32.25 Million/850 = \$37,940/space 100% = \$38,000 / space

Lowers Development Costs²

Overall development costs for a Robotic Parking System can be significantly lower than for a conventional garage. Since there are no people or running cars inside the garage, there are lower lighting and ventilation requirements. Lower insurance costs, lower personnel expenses and smaller land area required and more – all add up to increased savings.

Underground applications are particularly well suited to the Robotic Parking System Because only half the space is needed for the same

¹ "Robotic Parking System". 1994-2018. Retrieved from https://www.roboticparking.com/robotic_parking_increase_revenue.htm

² Ibid

amount of parking as conventional garages, developers can save as much as 50% on the excavation alone.

Operating and Capital Costs Conventional Garage vs Automated Parking Garage³

	Conventional Parking	Robotic Parking System
Capacity/ Labor Assumptions		
Capacity	892	892
Hours of Operation	24 / 7	24 / 7
Expenses		
Payroll & Benefits	\$850,000 (1)	\$145,000
Insurance Expenses	\$95,000	\$50,000
Utilities Expenses	\$165,000	\$200,000
Repairs & Maintenance	\$145,000	\$50,000
Bank Fee Expense	\$100,000	\$100,000
Marketing Expense	\$20,000	\$20,000

³ *Ibid*

Support Services Expense	\$75,000 (2)	\$35,000
Other Operating Expenses	\$150,000 (3)	\$75,000
Subtotal Operating Expenses	\$1,600,000	\$675,000
Real Estate Taxes Expense	\$150,000	\$150,000
Subtotal Non-Operating Expenses	\$150,000	\$150,000
Total Expenses	\$1,750,000	\$825,000
Capital Costs		
Security Camera / DVR System	\$30,000	\$30,000
Capital Account	\$240,000 (4)	\$60,000
Total Capital Costs	\$270,000	\$90,000
Grand Total	\$2,020,000	\$915,000

It's easy to see when you look at all the factors that a Robotic Parking System creates space for design and development and can help you increase revenues on your project.

(1) Labor Schedule

(2) Includes security, legal fees and audit fees

(3) Includes license/permit fees, uniforms, office supplies, claims, etc.

(4) Conventional Garage: 30% of garage repaired after 10 years at \$50/sf

Conclusion⁴

It's easy to see when you look at all the factors that a Robotic Parking System creates space for design and development and can help you increase revenues on your project⁵.

III. SOCIO-ECONOMIC STUDY

References:

1. Clegg S. (2017, April 19) The Four Project Parameters, Time, Cost, Quantity and Quality.
2. Mubashirin M. and Mahmud S. *Automated Car Parking System*. (A Thesis Submitted in Partial Fulfillment of the Requirement for the Degree of Bachelor of Science in Electrical and Electronic Engineering). Retrieved from http://dspace.bracu.ac.bd/xmlui/bitstream/handle/10361/8412/11121061%2C10121040_EEE.pdf?sequence=1&isAllowed=y
3. "Robotic Parking System". 1994-2018. Retrieved from https://www.roboticparking.com/robotic_parking_increase_revenue.htm

⁴ *Ibid*

